3.3 The Gaps in the Rationals

The rationals now have everything Section 3.2 asked of them: they form an ordered field (Theorem 3.2.1), every rational lies below a natural number (Theorem 3.2.3), and no two rationals are neighbors, since their midpoint lies between them. Is anything missing? School geometry suggests an answer. The diagonal of a square with side 1 has a length whose square is 2, and no fraction seems to measure it. That picture is motivation only. This section proves the two facts behind it from the construction: no rational squares to 2, and the set of rationals whose square is below 2 has no least upper bound in $\mathbb{Q}$. We also build a sequence of rationals that closes in on the gap from above without ever landing in it. A later chapter of this volume fills the gap.

3.3.1 No rational squares to two

Throughout, $x^2 \coloneqq x \cdot x$, and $2 \coloneqq 1 + 1$, $4 = 2 \cdot 2$ and so on are the natural numbers of these names, read in $\mathbb{Q}$ through the identifications of Sections 2.2 and 3.1; their sums, products and comparisons may be computed in $\mathbb{N}$ (Proposition 2.2.3, Proposition 3.1.4, Theorem 2.4.2 (a), Theorem 3.2.1 (b)).

Logic and Proof, Chapter 2 proved by parity that $\sqrt{2}$ is irrational (Logic and Proof, Theorem 2.2.1), for the school rationals. Parity is available for the constructed $\mathbb{Z}$ too (Corollary 2.5.3), and Exercise C1 allows it. Here we take a route that needs only the order and the least elements of $\mathbb{Z}$, and that adapts to other non-squares (Exercise C2).

Theorem 3.3.1. No Rational Squares to Two

There is no $x \in \mathbb{Q}$ with $x^2 = 2$.

Idea. Work backwards from a supposed square root $x = p/q$. This is the smallest-counterexample method (Sets and Functions, Example (Method: the smallest counterexample)) run on denominators, as in Sets and Functions, Lemma 3.2.4; here no lowest terms are needed, only a least denominator. From $x^2 = 2$ we get $2 - x = x^2 - x = x(x - 1)$, so

$$x = \frac{2 - x}{x - 1} = \frac{2 - p/q}{p/q - 1} = \frac{2q - p}{p - q}.$$

Since $1 < x < 2$ (as $1^2 < 2 < 2^2$), the new denominator $p - q$ is positive and smaller than q. So no fraction for x can have a **least** positive denominator, while Proposition 2.4.4 says some fraction must. The written proof checks the descent with integers only: it never divides, and it verifies $(2q - p)^2 = 2(p - q)^2$ by expanding. *Before reading on: using only integers, why is the new denominator $q' \coloneqq p - q$ positive, and why is $q' < q$?*

Proof. Suppose, for a contradiction, that $x \in \mathbb{Q}$ and $x^2 = 2$. Then $x \neq 0$, since $0^2 = 0$ (Proposition 2.3.2 (b)) and $2 > 0$. By (O1), $x > 0$ or $x < 0$; let $y \coloneqq x$ in the first case and $y \coloneqq -x$ in the second. Then $y > 0$ (Proposition 2.4.3 (a)) and $y^2 = x^2 = 2$, because $(-x)(-x) = xx$ (Proposition 2.3.2 (c)).

By Proposition 3.1.4 (b), $y = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, and $a > 0$ by Theorem 3.2.1 (a). By Definition 3.1.2, $y^2 = a^2/b^2$, and $2 = 2/1$, so $y^2 = 2$ says $a^2 \cdot 1 = 2b^2$ (Proposition 3.1.4

(b)). Hence the set

$$D := \{q \in \mathbb{Z}_{>0} : p^2 = 2q^2 \text{ for some } p \in \mathbb{Z}_{>0}\}$$

contains b, so it is non-empty, and 0 is a lower bound of it. By Proposition 2.4.4, D has a least element q. Choose $p \in \mathbb{Z}_{>0}$ with $p^2 = 2q^2$.

$q < p$. Suppose $p \leq q$. Since $p, q \geq 0$, Proposition 2.4.3 (g) gives $p^2 < q^2$ if $p < q$, and $p^2 = q^2$ if $p = q$; so $p^2 \leq q^2$, that is, $q^2 + q^2 \leq q^2$, and adding $-q^2$ (O3) gives $q^2 \leq 0$. But $q \neq 0$, so $q^2 > 0$ by Proposition 2.4.3 (e). This contradicts (O1), so $p \leq q$ fails, and $q < p$ by totality (Proposition 2.4.3 (i)).

$p < 2q$. Suppose $2q \leq p$. Since $2q, p \geq 0$, (g) (or equality, if $2q = p$) gives $(2q)^2 \leq p^2$, that is, $2q^2 + 2q^2 = 4q^2 \leq 2q^2$, so $2q^2 \leq 0$. But $2q^2 > 0$ by (O4). As before, $p < 2q$.

Descent. Let $p' := 2q - p$ and $q' := p - q$, integers. By Proposition 2.4.3 (a), $p < 2q$ gives $p' > 0$, and $q < p$ gives $q' > 0$; so $p', q' \in \mathbb{Z}_{>0}$ (Theorem 2.4.2 (b)). Adding $-q$ to $p < 2q$ (O3) gives $q' = p - q < q$. Expanding with Proposition 2.3.2 (c) and (d),

$$\begin{aligned} p'^2 - 2q'^2 &= (4q^2 - 4pq + p^2) - (2p^2 - 4pq + 2q^2) \\ &= 2q^2 - p^2 = 0. \end{aligned}$$

So $p'^2 = 2q'^2$ with $p' \in \mathbb{Z}_{>0}$, that is, $q' \in D$. But $q' < q$, and q is the least element of D. This contradiction shows that no $x \in \mathbb{Q}$ has $x^2 = 2$.

■

The answer to the prompt: $q' = p - q > 0$ because $q < p$, and $q' < q$ because $p < 2q$. These are the two inequalities "$1 < x < 2$" of the Idea, written for integers. The hypothesis $p^2 = 2q^2$ is used in each of the three steps, and the minimality of q only once, in the final contradiction.

Quick check. Run the descent on $p = 2$, $q = 1$, a pair with $p^2 = 4q^2$. Which step fails, and why must it?

Solution. The step $p < 2q$ fails: $2q \leq p$ now gives $4q^2 \leq p^2 = 4q^2$, no contradiction, and indeed $p = 2q$. Then $p' = 2q - p = 0$ is not positive and $q' = p - q = 1$ is not below q (and $p'^2 - 2q'^2 = -2$). A descent must fail here, since $2^2 = 4$.

3.3.2 The set below the gap

Sets and Functions, Section 5.3 stated, without proof, that the set of positive rationals with square below 2 has no least upper bound in $\mathbb{Q}$, and pointed out that the ambient set decides whether a supremum exists. We now prove it. By Proposition 2.4.3 (i) and Theorem 3.2.1, $\leq$ is a total order on $\mathbb{Q}$, so $(\mathbb{Q}, \leq)$ is a partially ordered set and the words of *Sets and Functions*, Section 5.3 apply: upper bounds (Sets and Functions, Definition 5.3.2) and a supremum, an upper bound s with (S1) $x \leq s$ for every x in the set and (S2) $s \leq u$ for every upper bound u (Sets and Functions, Definition 5.3.5).

Theorem 3.3.2. A Bounded Set of Rationals with No Rational Supremum

Let

$$A := \{x \in \mathbb{Q} : x^2 < 2\}, \qquad A^+ := \{x \in A : x > 0\} = \{x \in \mathbb{Q} : x > 0,\ x^2 < 2\}.$$

(a) A is non-empty and bounded above in $(\mathbb{Q}, \leq)$, but A has no supremum in $\mathbb{Q}$.

(b) A and A^+ have the same upper bounds in $\mathbb{Q}$. Hence A^+ is also non-empty and bounded above, and has no supremum in $\mathbb{Q}$.

Try it first (ten minutes): a rational $s > 0$ is offered as $\sup A$. Find a rational that proves the offer wrong, once in the case $s^2 < 2$ and once in the case $s^2 > 2$.

Idea (Scratch work). Suppose a rational s is the supremum. Unwind what that gives and what would refute it.

Givens	Goal
s is a supremum of A: (S1) and (S2)	a contradiction
$1 \in A$, so $s \geq 1 > 0$; $s^2 \neq 2$ by Theorem 3.3.1	split by trichotomy: $s^2 < 2$ or $s^2 > 2$
$s^2 < 2$	an element of A above s, against (S1)
$s^2 > 2$	an upper bound of A below s, against (S2)

If $s^2 < 2$, we need an element of A above s. The obvious try is $s + h$ with $h > 0$ small, but $(s+h)^2 = s^2 + 2sh + h^2$ has an h^2 in it, and taming that needs a separate size estimate on h. If $s^2 > 2$, we need a smaller upper bound, and a guess $s - h$ runs into the same algebra.

A cleaner idea finds the new rational in one formula. If $s^2 > 2$, then $2/s < s$ and $(2/s)^2 = 4/s^2 < 2$: the gap lies between $2/s$ and s. Their average

$$t(s) := \frac{1}{2}\left(s + \frac{2}{s}\right) = \frac{s^2 + 2}{2s}$$

is smaller than s, and its square is still above 2; that is the smaller upper bound. If $s^2 < 2$, apply the same trick from the other side: $t(s)$ overshoots, so $u(s) := 2/t(s) = 4s/(s^2+2)$ undershoots, and it lands above s. Each claim is an identity in the field $\mathbb{Q}$, checked by clearing denominators.

Proof.

(a) *Non-empty:* $1^2 = 1 < 2$, so $1 \in A$. *Bounded above:* 2 is an upper bound. Indeed, let $x \in A$, and suppose $x > 2$. Then $0 \leq 2 < x$, so $4 = 2^2 < x^2$ by Proposition 2.4.3 (g), and $2 < 4$, so $2 < x^2$, against $x^2 < 2$ by (O1). So $x \leq 2$, by totality.
Two maps. For $s \in \mathbb{Q}_{>0}$, we have $2s > 0$ by (O4), and $s^2 + 2 > 0$ by Proposition 2.4.3 (e) and (b); both are non-zero. Let

$$t(s) := \frac{s^2 + 2}{2s}, \qquad u(s) := \frac{4s}{s^2 + 2}.$$

Both are rationals, and both are positive, by Proposition 3.2.2 (a) and (O4). They satisfy

$$\begin{aligned} t(s)^2 - 2 &= \frac{(s^2-2)^2}{4s^2}, & s - t(s) &= \frac{s^2-2}{2s}, \\ u(s) - s &= \frac{s(2-s^2)}{s^2+2}, & 2 - u(s)^2 &= \frac{2(s^2-2)^2}{(s^2+2)^2}. \end{aligned} \tag{3.3.1}$$

Each identity is checked by clearing its non-zero denominator.

↻ **Second reading** · *Checking the four identities*

Multiply both sides of each identity by its denominator, which is non-zero, so that the multiplication can be undone with the inverse. For the first, $2s \cdot t(s) = s^2 + 2$, so $4s^2(t(s)^2-2) = (s^2+2)^2 - 8s^2 = s^4 - 4s^2 + 4 = (s^2-2)^2$. For the second, $2s(s-t(s)) = 2s^2 - (s^2+2) = s^2 - 2$. For the third, $(s^2+2)(u(s)-s) = 4s - s^3 - 2s = s(2-s^2)$. For the fourth, $(s^2+2)^2(2-u(s)^2) = 2(s^2+2)^2 - 16s^2 = 2(s^2-2)^2$.

No supremum. Suppose, for a contradiction, that $s \in \mathbb{Q}$ is a supremum of A. By (S1) and $1 \in A$, $s \geq 1 > 0$, so $s \in \mathbb{Q}_{>0}$. By Theorem 3.3.1, $s^2 \neq 2$, so by (O1), $s^2 < 2$ or $s^2 > 2$. In both cases $s^2 - 2 \neq 0$, so $(s^2-2)^2 > 0$ by Proposition 2.4.3 (e).
Case 1: $s^2 < 2$. Then $2 - s^2 > 0$, and the third identity of Eq. 3.3.1 exhibits $u(s) - s$ as a product of positive elements (with Proposition 3.2.2 (a) for the reciprocal), so $u(s) > s$. The fourth gives $2 - u(s)^2 > 0$, so $u(s)^2 < 2$ and $u(s) \in A$. So A has an element $u(s) > s$, and s is not an upper bound of A, against (S1).
Case 2: $s^2 > 2$. The first identity gives $t(s)^2 - 2 > 0$, so $t(s)^2 > 2$. Then $t(s)$ is an upper bound of A: if some $x \in A$ had $x > t(s)$, then $0 < t(s) < x$ would give $2 < t(s)^2 < x^2$ by Proposition 2.4.3 (g), against $x^2 < 2$; so $x \leq t(s)$ for every $x \in A$. By (S2), $s \leq t(s)$. But the second identity, with $s^2 - 2 > 0$ and $2s > 0$, gives $s - t(s) > 0$, that is, $t(s) < s$. Then $t(s) < s \leq t(s)$, so $t(s) < t(s)$, against (O1).
Both cases are impossible, so A has no supremum in $\mathbb{Q}$.

(b) Every upper bound of A bounds $A^+ \subseteq A$. Conversely, if u bounds A^+, then $u \geq 1 > 0$, since $1 \in A^+$; and an $x \in A$ is either positive, so $x \in A^+$ and $x \leq u$, or $x \leq 0 < u$. So A and A^+ have the same upper bounds. Conditions (S1) and (S2) mention only the set of upper bounds, so by (a), A^+ has no supremum. It contains 1 and is bounded above by 2.

■

Quick check. Compute $u(1/2)$ and $t(3)$, and say what each refutes.

Solution. $u(1/2) = 2/(9/4) = 8/9$, and $(8/9)^2 = 64/81 < 2$, so $8/9 \in A$ lies above $1/2$: $1/2$ is not an upper bound. $t(3) = 11/6$, and $(11/6)^2 = 121/36 > 2$ ($121 > 72$), so $11/6$ is an upper bound below 3: 3 is not the least.

The set A^+ is the set of *Sets and Functions*, Section 5.3, so (b) is the statement left unproved there. *Analysis on the Real Line*, Chapter 1 proves it again and then shows that A^+ has a supremum in the real numbers: that is the sense of "gap".

Warning. "A has no supremum in $\mathbb{Q}$, so A is unbounded" mixes up the two conditions a supremum needs. A is bounded above, by 2 for instance; what fails is (S2): among the rational upper bounds there is no least one, because every upper bound s is beaten by the smaller upper bound $t(s)$.

Quick check. In $\mathbb{Q}$, is $3/2$ the supremum of $\{x \in \mathbb{Q} : x^2 < 9/4\}$? Which step of the proof of (a) fails for this set?

Solution. Yes. (S1): $x > 3/2 > 0$ gives $x^2 > 9/4$ (Proposition 2.4.3 (g)), so every element of the set is at most $3/2$. (S2): if $v < 3/2$ bounded the set, let w be the larger of v and 1, and c the midpoint of w and $3/2$ (Section 3.2); then $0 < c < 3/2$, so $c^2 < 9/4$ and c is in the set, and $c > w \geq v$. The step that fails is "$s^2 \neq 2$": here $s^2 = 9/4$ is the target value, the third case of the trichotomy, which neither t nor u refutes.

3.3.3 A rational sequence aimed at the gap

The map t does more than refute a candidate. Started at an upper bound, it produces a smaller one, and it can be applied again. Iterating it gives a sequence of rationals that approaches the gap from above.

Example. A Rational Sequence Aimed at the Gap

For $s \in \mathbb{Q}_{>0}$ let $t(s) := (s^2 + 2)/(2s)$. Define $x_0 := 2$ and $x_{n+1} := t(x_n)$ for $n \in \mathbb{N}$, and let $e_n := x_n^2 - 2$. Show that this defines a sequence with $x_n \in \mathbb{Q}_{>0}$ for every $n \in \mathbb{N}$, and that for every $n \in \mathbb{N}$:

(i) $e_n > 0$, that is, $x_n^2 > 2$;
(ii) $e_{n+1} = e_n^2/(4x_n^2) \leq e_n^2/8$;
(iii) $x_n - x_{n+1} = e_n/(2x_n) > 0$, so $x_{n+1} < x_n$.

Compute $x_0, \ldots, x_3$ and $e_0, \ldots, e_3$.

Solution. The sequence. For $s \in \mathbb{Q}_{>0}$, $s^2 + 2 > 0$ and $2s > 0$, so $t(s) \in \mathbb{Q}_{>0}$ by (O4) and Proposition 3.2.2 (a): t maps $\mathbb{Q}_{>0}$ to $\mathbb{Q}_{>0}$. The recursion theorem (Sets and Functions, Theorem 3.3.1, which holds on $\mathbb{N}$ by Corollary 1.4.3 (b)), with $X = \mathbb{Q}_{>0}$, starting value 2 and $g(n, s) := t(s)$, gives exactly one function $n \mapsto x_n$ from $\mathbb{N}$ to $\mathbb{Q}_{>0}$ with $x_0 = 2$ and $x_{n+1} = t(x_n)$. So every x_n is a positive rational, with no further argument.

(i). We use Logic and Proof, Axiom 4.2.1 (on $\mathbb{N}$ by Corollary 1.4.3 (b)) for $P(n)$: $e_n > 0$. Base $P(0)$: $e_0 = 2^2 - 2 = 2 > 0$. Step: suppose $e_n > 0$. By the first identity of Eq. 3.3.1 with $s = x_n$,

$$e_{n+1} = t(x_n)^2 - 2 = \frac{e_n^2}{4x_n^2}.$$

Here $e_n^2 > 0$ by Proposition 2.4.3 (e), since $e_n \neq 0$, and $1/(4x_n^2) > 0$ by Proposition 3.2.2 (a); so $e_{n+1} > 0$ by (O4). This proves (i) for every n.

(ii). The equation was just shown, for every n. By (i), $x_n^2 > 2$, so $4x_n^2 > 4 \cdot 2 = 8$ by Proposition 2.4.3 (c). From $0 < 8 < 4x_n^2$, Proposition 3.2.2 (b) gives $1/(4x_n^2) < 1/8$, and multiplying by $e_n^2 > 0$ (Proposition 2.4.3 (c)) gives $e_{n+1} < e_n^2/8$; in particular $e_{n+1} \leq e_n^2/8$.

(iii). By the second identity of Eq. 3.3.1 with $s = x_n$, $x_n - x_{n+1} = e_n/(2x_n)$, a product of the positive elements e_n and $1/(2x_n)$; so it is positive, and $x_{n+1} < x_n$ by Proposition 2.4.3 (a).

The first terms. By the definition of t,

$$x_1 = \frac{4+2}{4} = \frac{3}{2}, \quad x_2 = \frac{9/4+2}{3} = \frac{17}{12}, \quad x_3 = \frac{289/144+2}{17/6} = \frac{577}{408}.$$

The errors are $e_0 = 2$, $e_1 = 9/4 - 2 = 1/4$, $e_2 = 289/144 - 2 = 1/144$ and $e_3 = 577^2/408^2 - 2 = 1/166464$, since $577^2 = 332929 = 2 \cdot 166464 + 1$ and $408^2 = 166464$. As a check on (ii): $e_1 = 2^2/(4 \cdot 4) = 1/4$, $e_2 = (1/16)/9 = 1/144$, and $e_3 = (1/144)^2/(4 \cdot 289/144) = 1/(144 \cdot 1156) = 1/166464$.

For orientation only (decimals are not part of this volume's toolkit), here are the terms in school decimals:

n	x_n	$e_n = x_n^2 - 2$	x_n, roughly
0	2	2	2
1	3/2	1/4	1.5
2	17/12	1/144	1.41667
3	577/408	1/166464	1.414216

The terms decrease, each is an upper bound of A (Case 2 of the proof above), and by (ii) the error is roughly squared at each step: $e_3 = 1/166464$. Yet no rational L can be "the number they approach": the squares would approach L^2, the errors force $L^2 = 2$, and Theorem 3.3.1 forbids that. This is a picture, not a proof, since limits in $\mathbb{Q}$ are not defined yet. Chapter 4 defines them, with $\varepsilon \in \mathbb{Q}_{>0}$, and proves that this sequence is Cauchy in $\mathbb{Q}$ with no limit in $\mathbb{Q}$, citing (i)–(iii).

Quick check. Is $3/2$ an upper bound of $A = \{x \in \mathbb{Q} : x^2 < 2\}$? Is $17/12$? Is $4/3$?

Solution. The first two are. Each is positive, and $(3/2)^2 = 9/4 > 2$ because $9 > 8$, while $(17/12)^2 = 289/144 > 2$ because $289 > 288$ (the comparison formula, Theorem 3.2.1 (a)). A positive rational b with $b^2 > 2$ bounds A: if $x \in A$ had $x > b$, then $x^2 > b^2 > 2$ by Proposition 2.4.3 (g), against $x^2 < 2$.

But $4/3$ is not, although it is positive: $(4/3)^2 = 16/9 < 2$ because $16 < 18$, so the hypothesis $b^2 > 2$ fails, and indeed $7/5 \in A$, as $(7/5)^2 = 49/25 < 2$ ($49 < 50$), while $7/5 > 4/3$ ($21 > 20$).

Chapters 2 and 3 built two number systems by one method. From $\mathbb{N}$ to $\mathbb{Z}$: pairs (a, b), identified when $a + d = b + c$, so that subtraction becomes possible. From $\mathbb{Z}$ to $\mathbb{Q}$: pairs (p, q) with $q > 0$,

identified when $ps = rq$, so that division becomes possible. Chapter 5 takes the third step, from $\mathbb{Q}$ to the real numbers, by the same route: the representatives are sequences of rationals like the one above, identified when their difference tends to 0.

3.3.4 Exercises

3.3.4.1 A. Check your understanding

Exercise. A1

These questions are about the proof of Theorem 3.3.2.

(a) Where is Theorem 3.3.1 used, and what would go wrong without it?

(b) Run the proof on $s = 3/2$ and on $s = 7/5$: which case applies, which rational does the proof produce, and why does it refute s?

(c) Let F be an ordered field containing an element $r > 0$ with $r^2 = 2$, and let $A_F := \{x \in F : x^2 < 2\}$. Which step of the proof fails for A_F, and what happens to $t(r)$ and $u(r)$?

(d) State the idea of the proof in two sentences.

Solution.

(a) Once, right after a supposed supremum s is fixed: it excludes $s^2 = 2$, so that trichotomy leaves the two cases $s^2 < 2$ and $s^2 > 2$. Without it there is a third case, in which neither map helps: $(s^2 - 2)^2 = 0$, so the identities give $u(s) = s$ and $t(s) = s$.

(b) For $s = 3/2$: $s^2 = 9/4 > 2$ (as $9 > 8$), so Case 2. The proof produces $t(3/2) = (9/4 + 2)/3 = 17/12$, an upper bound of A, since $(17/12)^2 = 289/144 > 2$; and $17/12 < 3/2$, since $17 \cdot 2 = 34 < 36 = 3 \cdot 12$. So $3/2$ is not the **least** upper bound. For $s = 7/5$: $s^2 = 49/25 < 2$ (as $49 < 50$), so Case 1. The proof produces

$$u(7/5) = \frac{28/5}{49/25 + 2} = \frac{28/5}{99/25} = \frac{140}{99},$$

which lies in A, since $140^2 = 19600 < 19602 = 2 \cdot 99^2$, and exceeds $7/5$, since $140 \cdot 5 = 700 > 693 = 7 \cdot 99$. So $7/5$ is not an upper bound.

(c) The step "$s^2 \neq 2$": in F the candidate $s = r$ satisfies $r^2 = 2$, and it falls in neither case. Indeed r is the supremum of A_F in F: it is an upper bound (if $x > r > 0$, then $x^2 > 2$), and an upper bound $v < r$ would have $v > 0$ (as $1 \in A_F$) and $v^2 < 2$, and then $u(v) \in A_F$ with $u(v) > v$, as in Case 1. The maps fix r: $t(r) = (2 + 2)/(2r) = 2/r = r$, since $r \cdot r = 2$, and $u(r) = 4r/4 = r$.

(d) Suppose a rational s were the supremum; since no rational squares to 2, either $s^2 < 2$ or $s^2 > 2$. The explicit rationals $u(s)$ and $t(s)$ then give, respectively, an element of the set above s or an upper bound below s, so s fails (S1) or (S2).

3.3.4.2 B. Practice

Exercise. B1: Put the proof in order

Here are lines of a proof of Theorem 3.3.1, shuffled, together with two lines that do not belong. Write the lines of a correct proof in a correct order, and say what is wrong with each

line you leave out.

(i) Set $p' \coloneqq 2q - p$ and $q' \coloneqq p - q$; both are positive integers, by the two inequalities between p and q.

(ii) Write $x = a/b$ with $a, b \in \mathbb{Z}_{>0}$; then $a^2 = 2b^2$.

(iii) Since $p^2 = 2q^2$, we have $p < q$.

(iv) Since $q' = p - q < q$, this contradicts the choice of q; so no rational squares to 2.

(v) Suppose $x \in \mathbb{Q}$ has $x^2 = 2$; replacing x by $-x$ if necessary, take $x > 0$.

(vi) $p < 2q$, since $p \geq 2q$ would give $p^2 \geq 4q^2 > 2q^2$.

(vii) Choose such an x with the least positive value.

(viii) Among all pairs (p, q) of positive integers with $p^2 = 2q^2$, choose one with q least (Proposition 2.4.4; b is such a q).

(ix) Expanding, $p'^2 - 2q'^2 = 2q^2 - p^2 = 0$, so $p'^2 = 2q'^2$.

(x) $q < p$, since $p \leq q$ would give $2q^2 = p^2 \leq q^2$, that is, $q^2 \leq 0$.

Solution. An admissible order is

$$\text{(v)},\ \text{(ii)},\ \text{(viii)},\ \text{(x)},\ \text{(vi)},\ \text{(i)},\ \text{(ix)},\ \text{(iv)}.$$

Lines (x) and (vi) may be exchanged, since neither uses the other. Otherwise the order is forced by what each line uses: (ii) needs the $x > 0$ of (v); (viii) needs the pair (a, b) of (ii), which makes the set of such q non-empty; (x) and (vi) are about the pair fixed in (viii), and if they came before it, the pair might change afterwards; (i) needs both inequalities; (ix) needs the definitions in (i); and (iv) needs $q' \in \mathbb{Z}_{>0}$ with $p'^2 = 2q'^2$, from (i) and (ix), and $q' < q$, which comes from (vi).

Line (vii) is legal but does no work. There is only one positive rational x with $x^2 = 2$: if $0 < x < y$ had $x^2 = y^2 = 2$, then Proposition 2.4.3 (g) would give $2 = x^2 < y^2 = 2$, against (O1). So "the least" just picks out the x of (v) and gives nothing new. The descent does not produce a smaller x: it keeps the number and shrinks its denominator, $p'/q' = p/q$, so choosing x least never leads to a contradiction. What must be least is the denominator q, as in (viii).

Line (iii) fails. Its conclusion is false: from $p^2 = 2q^2$ and $q > 0$ we get $p^2 > q^2$, hence $q < p$ by Proposition 2.4.3 (g), the opposite of (iii). It also clashes with (x).

Exercise. B2: Prove or disprove

Prove or disprove each statement.

(a) There is no $x \in \mathbb{Q}$ with $x^2 = 8$.

(b) For all $x, y \in \mathbb{Q}$, if $x^2 < y^2$, then $x < y$.

(c) If $s \in \mathbb{Q}$, $s > 0$ and $s^2 > 2$, then $(2/s)^2 < 2$.

(d) If $s \in \mathbb{Q}$, $s > 0$ and $s^2 < 2$, then $(s + 1/10)^2 < 2$.

Solution.

(a) **True.** If $x^2 = 8$, then $y := x/2$ has $y^2 = x^2/4 = 2$ (Definition 3.1.2), against Theorem 3.3.1.

(b) **False:** $x = 1$, $y = -2$ give $x^2 = 1 < 4 = y^2$, but $1 > -2$. (Rule (g) of Proposition 2.4.3 needs $0 \le x$ and $0 \le y$.)

(c) **True.** $(2/s)^2 = 4/s^2$. From $s^2 > 2 > 0$, Proposition 3.2.2 (b) gives $1/s^2 < 1/2$, and multiplying by $4 > 0$ (Proposition 2.4.3 (c)) gives $4/s^2 < 2$. (So the map $s \mapsto 2/s$ carries upper bounds of A^+ into A^+; the proof's $u(s)$ improves on it.)

(d) **False:** $s = 7/5$ has $s^2 = 49/25 < 2$ ($49 < 50$), but $s + 1/10 = 3/2$ and $(3/2)^2 = 9/4 > 2$ ($9 > 8$). A fixed step can overshoot; the proof's $u(s)$ is chosen from s.

Exercise. B3

Compute x_4 of Example (A Rational Sequence Aimed at the Gap), as a fraction of integers, and $e_4 = x_4^2 - 2$. Check your value of e_4 against (ii) of the example. Hence x_4 is an upper bound of $\{x \in \mathbb{Q} : x^2 < 2\}$ below x_3.

Solution. With $x_3 = 577/408$,

$$x_4 = t(x_3) = \frac{577^2 + 2 \cdot 408^2}{2 \cdot 577 \cdot 408} = \frac{332929 + 332928}{470832} = \frac{665857}{470832}.$$

Then $665857^2 = 443365544449$ and $470832^2 = 221682772224$, and $443365544449 - 2 \cdot 221682772224 = 1$, so

$$e_4 = \frac{1}{221682772224}.$$

By (ii), $e_4 = e_3^2/(4x_3^2) = 1/(166464 \cdot 4 \cdot 332929)$, and $4 \cdot 166464 \cdot 332929 = 221682772224$, which agrees. Since $x_4 > 0$ and $e_4 > 0$, x_4 bounds the set (the argument of the Quick check after the table), and $x_4 < x_3$ by (iii).

3.3.4.3 C. Going deeper

Exercise. C1: Exit check: the gap at $\sqrt{2}$

Prove that no rational number squares to 2, and that the set $\{x \in \mathbb{Q} : x^2 < 2\}$ has no supremum in $\mathbb{Q}$.

Solution. No rational squares to 2. Suppose $x \in \mathbb{Q}$ with $x^2 = 2$. Then $x \neq 0$, and replacing x by $-x$ if needed ($(-x)^2 = x^2$), $x > 0$. Write $x = a/b$ with $a, b \in \mathbb{Z}_{>0}$ (Proposition 3.1.4 (b), Theorem 3.2.1 (a)); then $a^2 = 2b^2$. Among all $q \in \mathbb{Z}_{>0}$ for which some $p \in \mathbb{Z}_{>0}$ has $p^2 = 2q^2$, take the least (Proposition 2.4.4; the set contains b and is bounded below by 0), with its p. Then $q < p$ (else $p^2 \le q^2$ by Proposition 2.4.3 (g), so $q^2 \le 0$, against $q^2 > 0$) and $p < 2q$ (else $4q^2 \le p^2 = 2q^2$, so $2q^2 \le 0$). So $p' := 2q - p$ and $q' := p - q$ are positive integers with $q' < q$, and $p'^2 - 2q'^2 = 2q^2 - p^2 = 0$ by expanding. This contradicts the choice of q.

Bounded above. 2 bounds the set A: $x > 2$ would give $x^2 > 4 > 2$ (Proposition 2.4.3 (g)). And $1 \in A$.

No supremum. Suppose $s \in \mathbb{Q}$ is a supremum of A. As $1 \in A$, $s \geq 1 > 0$, and $s^2 \neq 2$ by the first part.

If $s^2 < 2$, let $u := 4s/(s^2+2) \in \mathbb{Q}_{>0}$. By the third and fourth identities of Eq. 3.3.1, $u - s > 0$ and $2 - u^2 > 0$. So $u \in A$ and $u > s$: s is not an upper bound.

If $s^2 > 2$, let $t := (s^2+2)/(2s) \in \mathbb{Q}_{>0}$. By the first and second identities of Eq. 3.3.1, $t^2 - 2 > 0$ and $s - t > 0$. So $t^2 > 2$, and t bounds A (an $x \in A$ with $x > t > 0$ would have $x^2 > t^2 > 2$); and $t < s$. So s is not the least upper bound.

By trichotomy one of the two cases holds, and each contradicts the choice of s. So A has no supremum in $\mathbb{Q}$.

Marking scheme. (10 points) - 4: no rational squares to 2: a least denominator, or another correct route (1); $q < p < 2q$, or the parity step (2); the contradiction (1). - 1: shows $1 \in A$, so that a supremum s satisfies $s \geq 1 > 0$ (an upper bound of A is optional). - 2: the case $s^2 < 2$: an explicit element of A larger than s, with both properties verified. - 2: the case $s^2 > 2$: an explicit smaller rational, verified to be an upper bound of A. - 1: concludes by trichotomy, with $s^2 = 2$ excluded by the first part. Fatal errors (0 for the step): assuming that $\sqrt{2}$ exists (in $\mathbb{Q}$ or in a larger field) and comparing s with it; "$s^2 < 2$, so $s + \varepsilon \in A$ for small ε" with no ε exhibited and checked. A correct parity route (every integer is $2m$ or $2m+1$, and not both, Corollary 2.5.3 (a)) earns the same 4 points.

Exercise. C2: Which integers have a rational square root?

For which $n \in \mathbb{Z}_{>0}$ does the set $A_n := \{x \in \mathbb{Q} : x^2 < n\}$ have a supremum in $\mathbb{Q}$? Prove your answer.

Solution (Hint 1). First find $k \in \mathbb{N}$ with $k^2 \leq n < (k+1)^2$, and treat $n = k^2$ on its own. For $k^2 < n$, adapt both proofs of this section: the descent, and then the two maps.

Solution (Hint 2). Let $T := \{m \in \mathbb{Z} : m \geq 0,\ n < m^2\}$. It contains $n+1$, it is bounded below by 0, and $0 \notin T$; so its least element is $k+1$ for some $k \in \mathbb{N}$. In the descent for $k^2 < n$, try $q' := p - kq$.

Solution. **Answer:** exactly when $n = k^2$ for some $k \in \mathbb{N}$, and then the supremum is k.

Finding k. Let $T := \{m \in \mathbb{Z} : m \geq 0,\ n < m^2\}$. Since $n+1 \geq 1$, $(n+1)^2 \geq n+1 > n$ (Proposition 2.4.3 (c)), so $n+1 \in T$; 0 is a lower bound of T; and $0 \notin T$, as $0^2 = 0 < n$. By Proposition 2.4.4, T has a least element l. Since $0 \notin T$ and $l \geq 0$, we have $l > 0$, so $l \geq 1$ by discreteness (Theorem 2.4.2 (c)); then $l - 1 \geq 0$ by (O3), so $k := l - 1 \in \mathbb{N}$ by Theorem 2.4.2 (b). Then $n < (k+1)^2$, and $k \notin T$ with $k \geq 0$ gives $k^2 \leq n$. So either $n = k^2$ or $k^2 < n < (k+1)^2$.

Case $n = k^2$. Here $k \geq 1$, since $n \geq 1$. We show $k = \sup A_n$. (S1): if $x \in A_n$ had $x > k$, then $x > k \geq 0$ would give $x^2 > k^2 = n$ (Proposition 2.4.3 (g)), against $x^2 < n$; so $x \leq k$. (S2): let v be an upper bound of A_n, and suppose $v < k$. Let $w := v$ if $v \geq 0$ and $w := 0$

otherwise; then $v \leq w$ and $0 \leq w < k$. Let c be the midpoint of w and k (Section 3.2): $w < c < k$. Then $c > 0$ and $c^2 < k^2 = n$ by (g), so $c \in A_n$, and $c > w \geq v$, so v is not an upper bound. Hence $k \leq v$ for every upper bound v, and k is the supremum.

Case $k^2 < n < (k+1)^2$. Then $n \geq 2$, since $n = 1 = 1^2$ is the other case; so $1 \in A_n$.

(i) *No rational squares to* n. Suppose $x^2 = n$ with $x \in \mathbb{Q}$; as in Theorem 3.3.1, we may take $x > 0$, and then $p^2 = nq^2$ for some $p, q \in \mathbb{Z}_{>0}$, with q least among all such denominators (Proposition 2.4.4). Multiplying $k^2 < n < (k+1)^2$ by $q^2 > 0$ gives $(kq)^2 < p^2 < ((k+1)q)^2$, so $kq < p < (k+1)q$ by (g). Let

$$p' \coloneqq nq - kp, \qquad q' \coloneqq p - kq.$$

Then $0 < q' < q$. For p', expand:

$$q \cdot p' = nq^2 - kpq = p^2 - kpq = p(p - kq) > 0$$

by (O4); since $q > 0$, $p' \leq 0$ would give $qp' \leq 0$ (Proposition 2.4.3 (c)), so $p' > 0$. Finally, expanding,

$$p'^2 - nq'^2 = (k^2 - n)(p^2 - nq^2) = 0,$$

so q' is a smaller admissible denominator, a contradiction.

(ii) *No supremum.* The proof of Theorem 3.3.2 (a) goes through with n in place of 2, using (i) where it used Theorem 3.3.1, and A_n contains 1 and is bounded above by n. For $s \in \mathbb{Q}_{>0}$, let

$$t(s) \coloneqq \frac{s^2 + n}{2s}, \qquad u(s) \coloneqq \frac{2ns}{s^2 + n}.$$

Clearing denominators as before gives the four identities

$$t(s)^2 - n = \frac{(s^2 - n)^2}{4s^2}, \qquad s - t(s) = \frac{s^2 - n}{2s},$$
$$u(s) - s = \frac{s(n - s^2)}{s^2 + n}, \quad n - u(s)^2 = \frac{n(s^2 - n)^2}{(s^2 + n)^2}.$$

So A_n has a supremum in $\mathbb{Q}$ exactly when n is the square of a natural number.

Marking scheme. (10 points) - 2: produces $k \in \mathbb{N}$ with $k^2 \leq n < (k+1)^2$, from a least element. - 2: the case $n = k^2$: checks (S1) and (S2) for k. - 3: the case $k^2 < n$: no rational squares to n, by a descent with a least denominator (or another correct route), including a proof that the new numerator is positive. - 2: the two maps (or any explicit rationals) and the two cases, as in the proof for 2. - 1: states the answer: exactly the squares of natural numbers. Fatal errors (0 for the step): using a square root of n, or the floor of one, as if it existed; deducing $nq - kp > 0$ from $kq < p < (k+1)q$ alone, which fails for $n = 5$, $k = 2$, $q = 10$, $p = 28$.

3.3.4.4 R. Review

Exercise. R1

Recall that an integer is a class $[(a, b)]$ of pairs of natural numbers, with $(a, b) \sim (c, d)$ if and only if $a + d = b + c$. For $(a, b) \in \mathbb{N} \times \mathbb{N}$, let $f(a, b) \coloneqq (a, b)$ if $b \leq a$, and $f(a, b) \coloneqq (b, a)$ if

$a < b$. Does $[(a, b)] \mapsto [f(a, b)]$ define a function $\mathbb{Z} \to \mathbb{Z}$? Prove your answer, and say which integer it assigns to $x \geq 0$ and to $x < 0$.

Solution. **Yes.** By Theorem 2.2.1 (a), with $X = Y = \mathbb{N} \times \mathbb{N}$, it suffices to show that $(a, b) \sim (c, d)$ implies $f(a, b) \sim f(c, d)$ (the representative check, Sets and Functions, Example (Method: checking that a definition on classes is well defined)). Let $a + d = b + c$, and $x \coloneqq [(a, b)] = [(c, d)]$.

The cases match. $x \in P_{\mathbb{Z}}$ if and only if $b < a$, whichever representative is used (the well-definedness paragraph after Definition 2.4.2), and $x = 0$ if and only if $a = b$, since $(a, b) \sim (0, 0)$ means $a = b$. So $b \leq a \iff (x \in P_{\mathbb{Z}}$ or $x = 0) \iff d \leq c$.

Hence either $b \leq a$ and $d \leq c$, and then $f(a, b) = (a, b) \sim (c, d) = f(c, d)$; or $a < b$ and $c < d$, and then $f(a, b) = (b, a)$, $f(c, d) = (d, c)$, and $b + c = a + d$ says $(b, a) \sim (d, c)$.

Since $x \geq 0$ if and only if $b \leq a$, and $-[(a, b)] = [(b, a)]$ (Definition 2.2.1), the function sends x to x if $x \geq 0$ and to $-x$ if $x < 0$: it is the familiar $x \mapsto |x|$.

Exercise. R2

Show that the set $\mathbb{Q}$, as constructed in this chapter, is countable.

Solution. The tools are the results on countable sets of *Sets and Functions*, Chapter 4, which run on $\mathbb{N}$ by the citing rule after Corollary 1.4.3 (group (i) for Sets and Functions, Proposition 4.2.3 and Sets and Functions, Proposition 4.2.4; group (ii), citable in review exercises, for Sets and Functions, Theorem 4.2.6 and Sets and Functions, Corollary 4.2.7 (a)).

$\mathbb{Z}$ is countable. $\mathbb{N} \times \mathbb{N}$ is countable (Sets and Functions, Theorem 4.2.6). The map $a \mapsto [a]$, from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\!\sim$, is surjective, since every class is the class of each of its elements (Sets and Functions, Definition 5.2.4). By Sets and Functions, Proposition 4.2.4, $\mathbb{Z}$ is countable.

$\mathbb{Z} \times \mathbb{Z}_{>0}$ is countable. $\mathbb{Z}_{>0} \subseteq \mathbb{Z}$ is countable by Sets and Functions, Proposition 4.2.3, and the product of two countable sets is countable (Sets and Functions, Corollary 4.2.7 (a)).

$\mathbb{Q}$ is countable. The map $(p, q) \mapsto [(p, q)]$, from $\mathbb{Z} \times \mathbb{Z}_{>0}$ to $\mathbb{Q}$, is surjective (Sets and Functions, Definition 5.2.4), so $\mathbb{Q}$ is countable by Sets and Functions, Proposition 4.2.4.

The conclusion is Sets and Functions, Theorem 4.2.8 for the constructed $\mathbb{Q}$; that theorem, about the school rationals, is not used.

Exercise. R3

Let $A \coloneqq \{x \in \mathbb{Q} : x^2 < 2\}$ and $s \in \mathbb{Q}$. Write "s is a supremum of A in $\mathbb{Q}$" in symbols, with bounded quantifiers. Then write its negation, moving $\neg$ inward as far as it goes and naming the law used at each step. The proof that A has no supremum in $\mathbb{Q}$ has two cases; which half of the negation does each case establish, and with which witness?

Solution. By Sets and Functions, Definition 5.3.5, "s is a supremum of A" is the conjunction of (S1) and (S2):

$$(\forall x \in A \;\; x \leq s) \;\wedge\; \forall v \in \mathbb{Q}\; ((\forall x \in A \;\; x \leq v) \to s \leq v).$$

Its negation is, by Logic and Proof, Proposition 2.5.5 (d),

$$\neg(\forall x \in A \;\; x \leq s) \;\vee\; \neg\forall v \in \mathbb{Q}\; ((\forall x \in A \;\; x \leq v) \to s \leq v).$$

By Logic and Proof, Proposition 6.2.3 (a), the first disjunct becomes $\exists x \in A\; \neg(x \leq s)$, that is, $\exists x \in A \;\; s < x$ (totality of $\leq$, Proposition 2.4.3 (i)). By Logic and Proof, Proposition 6.2.3 (a) and then Logic and Proof, Proposition 6.2.2, the second becomes

$$\exists v \in \mathbb{Q}\; ((\forall x \in A \;\; x \leq v) \wedge \neg(s \leq v)),$$

that is, there is an upper bound v of A with $v < s$. In words: s fails to be the supremum when **some element of A lies above s**, or **some upper bound lies below s**.

In the proof of Theorem 3.3.2, Case 1 ($s^2 < 2$) establishes the first disjunct, with witness $x = u(s) = 4s/(s^2+2)$. Case 2 ($s^2 > 2$) establishes the second, with witness $v = t(s) = (s^2+2)/(2s)$. (The proof itself needs no negation law: it derives a contradiction from (S1) in Case 1 and from (S2) in Case 2.)

Exercise. R4

Is there a rational number x with $x^3 = 2$, where $x^3 := x^2 \cdot x$? Prove your answer.

Solution. **No.** The tool is descent on the denominator, as in Theorem 3.3.1, run with parity.

Parity. Every integer is $2m$ or $2m+1$ for some $m \in \mathbb{Z}$, and not both (Corollary 2.5.3 (a)). An odd cube is odd:

$$(2m+1)^3 = 2(4m^3 + 6m^2 + 3m) + 1.$$

Descent. Suppose $x \in \mathbb{Q}$ and $x^3 = 2$. Then $x > 0$: $x = 0$ gives $x^3 = 0$, and $x < 0$ gives $x^2 > 0$ and $x^3 = x^2 \cdot x < 0$ (Proposition 2.4.3 (e), (c)). So $x = p/q$ with $p, q \in \mathbb{Z}_{>0}$ and $p^3 = 2q^3$ (Proposition 3.1.4 (b), Definition 3.1.2). Take q least among such denominators (Proposition 2.4.4), with its p. Since $p^3 = 2q^3$ has the even form, p is not odd (its cube would be odd), so $p = 2m$ with $m \in \mathbb{Z}$, and $m > 0$ because $p > 0$ ($m \leq 0$ would give $2m \leq 0$). Then $2q^3 = 8m^3$, and canceling the non-zero factor 2 (Proposition 2.4.3 (f)), $q^3 = 2(2m^3)$. So q is not odd either: $q = 2l$ with $l \in \mathbb{Z}_{>0}$. Then $8m^3 = 2 \cdot 8l^3$, so $m^3 = 2l^3$, and $l < l + l = q$. So l is a smaller admissible denominator, with $m \in \mathbb{Z}_{>0}$: a contradiction.

Why the look-alike fails. The substitution $p' := 2q - p$, $q' := p - q$ of the square-root proof came from $x = (2-x)/(x-1)$, which used $x^2 = 2$. For cubes the same algebra fails: expanding,

$$(2q-p)^3 - 2(p-q)^3 = -3p^3 + 12p^2q - 18pq^2 + 10q^3,$$

which is not a constant multiple of $p^3 - 2q^3$ (it has a p^2q term), so this identity gives no way to derive $p'^3 = 2q'^3$ from $p^3 = 2q^3$, and the descent has no smaller pair to move to. Parity, by contrast, works for every power: "even cube, so even base" is the same argument for squares and cubes.